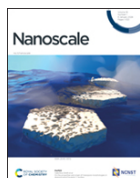




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Activated carbon adsorption enhanced removal of organic contaminants using a catalytic ceramic membrane with peroxymonosulfate activation

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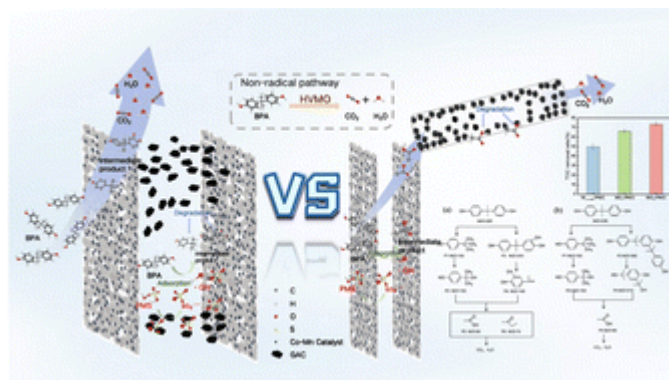
Abstract

Granular activated carbon (GAC) was respectively loaded into catalytic ceramic membrane channels (MG₁) and connected at the rear end of the catalytic ceramic membrane (MG₂) to improve the removal of emerging contaminants *via* peroxymonosulfate (PMS) activation. Compared to the catalytic ceramic membrane system (M_{Co-Mn}/PMS) without GAC coupling, the MG₁/PMS and MG₂/PMS systems demonstrated complete bisphenol A (BPA) removal within 5 min, with total organic carbon (TOC) removal rates increasing to 66% and 73%, respectively, and exhibited exceptional stability. Quenching

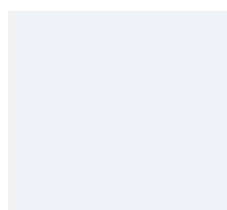
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1. Introduction

Membrane separation technologies, particularly ultrafiltration (UF) membranes, have become indispensable in potable water production and wastewater reclamation.^{1,2} Ceramic membranes, as inorganic ultrafiltration membranes, have a long service life and exhibit superior thermal stability, chemical resistance, and mechanical strength, making them ideal for organic wastewater

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limited efficacy against recalcitrant emerging pollutants.^{9–11} The integration of advanced oxidation processes (AOPs) with ceramic membrane filtration has consequently gained significant research traction. This technology generates multiple reactive oxygen species ($\cdot\text{OH}$, $\text{SO}_4^{\cdot-}$, and $^1\text{O}_2$) through heterogeneous activation pathways, achieving remarkable degradation rates over a wide pH range.^{12–14} Yu *et al.* prepared an optimized spinel-incorporated catalytic poly(vinylidene fluoride) membrane (spinel-PVDF), which effectively demonstrated the degradation of a wide range of pollutants. The membrane demonstrated high stability, maintaining over 95% degradation efficiency for 120 h at a flux of $100\text{ L m}^{-2}\text{ h}^{-1}$ (LMH).¹⁵ Zong *et al.* synthesized an FeOCl-modified carbon-based catalytic membrane (CM-FeOCl), which could effectively activate 80% of H_2O_2 under neutral conditions, and as a result, a high tetracycline (TC) removal efficiency exceeding 90% could be attained. The authors observed a 45% reduction in the total organic carbon (TOC) removal rate.¹⁶ Wu *et al.* developed a new catalytic system based on exfoliated MXene-modified BiFeO_3 (MX-BFO) for efficient degradation of antibiotics. Within 30 min, a 50 mg L^{-1} TC solution could be completely degraded in the system with a TOC removal rate of 54.5%.¹⁷ In addition, other excellent studies have discussed coupling methods between ceramic membranes and peroxymonosulfate (PMS)-based AOPs and their treatment of pollutants.^{18–21} Researchers generally believe that the coupling of ceramic membrane filtration and PMS will become a transformative technology for next-generation water treatment systems to address emerging pollutants.

Although the combination of AOPs and membrane filtration technology provides significant advantages, including strong mechanical strength, excellent chemical stability, and resistance to organic pollution under long-term oxidation conditions, achieving complete mineralization of contaminants remains a daunting challenge. Current research predominantly focuses on single catalytic membrane AOP systems for the removal of emerging pollutants, frequently neglecting comprehensive organic matter elimination, as evidenced by persistent TOC levels in treated effluents.^{22–24} Conventional strategies for efficient TOC reduction, including nanofiltration (NF), reverse osmosis (RO), electrochemical processes and AOPs, typically require substantial energy input and involve intricate operational protocols.^{25,26} These limitations highlight the need for more sustainable and efficient mineralization strategies. Based on our previous work, the single catalytic

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In this study, catalytic ceramic membranes were prepared by a facile impregnation–sintering method, and two distinct membrane modules were engineered by varying the positions of the GAC. A comparative investigation was conducted to evaluate the catalytic efficacy of the M_{Co-Mn}/PMS , MG_1/PMS and MG_2/PMS systems. The investigation systematically evaluated the TOC removal performance and PMS activation efficiency of these three systems. Simultaneously, the stability and reusability of the two GAC-integrated systems were also examined. The main active species involved in BPA degradation in the MG_1/PMS and MG_2/PMS systems were identified, and the degradation pathways were elucidated and toxicity analyses performed. Finally, the antifouling performance of M_{Co-Mn} was evaluated, and the main causes of membrane fouling were analyzed.

2. Materials and methods

2.1. Chemicals and reagents

Cobaltous nitrate hexahydrate ($Co(NO_3)_2 \cdot 6H_2O$), methanol (MeOH), *tert*-butanol (TBA), and dimethyl sulfoxide (DMSO) were purchased from Sinopharm Chemical Reagent Co., Ltd (Shanghai, China). Bisphenol A (BPA) was purchased from Aladdin Biochemical Co., Ltd (Shanghai, China). L-Histidine, *p*-benzoquinone (BQ), and peroxymonosulfate (PMS, $KHSO_5 \cdot 0.5KHSO_4 \cdot 0.5K_2SO_4$) were purchased from Shanghai Macklin Biochemical Co., Ltd (Shanghai, China). Unless otherwise stated, all chemicals employed in this study were of analytical grade and were not subjected to further purification. Deionized (DI) water was prepared using ultrapure water acquired from a Millipore Milli-Q water system. Raw water from the Dingjiahe Reservoir in Qingdao (China) was used to investigate membrane fouling and was subjected to preliminary treatment using a 0.45 μm membrane. The raw water quality parameters of Dingjiahe reservoir are shown in Table S1. The initial pH of the reaction solution was adjusted to 7 ± 0.2 , and the temperature was maintained at 25 ± 0.5 °C. Granular activated carbon (GAC) was purchased from Xinzhiyuan Activated Carbon Co., Ltd (China). The specific surface area and pore size distribution parameters of the GAC material are provided in Fig. S1 and Table S2. The flat-sheet ceramic membrane prepared in our laboratory was employed in this study, and its characteristic parameters are provided in Table S3.

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temperature, the catalytic ceramic membrane loaded with the Co–Mn catalyst was obtained and denoted as $M_{\text{Co–Mn}}$.

Two integrated catalytic ceramic membrane-activated carbon systems were constructed in this study. In one system, 5 g of GAC was loaded into the $M_{\text{Co–Mn}}$ membrane channels by encapsulation with epoxy resin, and the effective catalytic membrane area was $\sim 90 \text{ cm}^2$, which was then assembled with an immersion ceramic membrane filtration device and designated as the MG_1/PMS system. In another system, an activated carbon pipe was prepared using a 1 cm diameter polypropylene hose filled with 5 g of GAC, secured with a 1 cm^3 screen at both ends of the pipe and connected to the $M_{\text{Co–Mn}}$ membrane by a quick coupling. This integrated system was named the MG_2/PMS system. The specific physical devices of the two systems are shown in [Fig. 1](#).



Fig. 1 Schematic diagram of the two integrated catalytic ceramic membrane-activated carbon systems.

2.3. Experimental procedures

First, 200 mg L^{-1} of BPA solution was prepared by mixing 0.2 g of BPA powder in an appropriate amount of deionized water in a 1000 mL volumetric flask and diluting to a constant volume of 1000 mL.

As shown in [Fig. 1](#), the submerged ceramic membrane filtration experiment was carried out in a 1 L feed tank with 10 mg L^{-1} of BPA solution. The initial solution pH was adjusted with 0.1 M HCl and 0.1 M NaOH. After adding PMS to the feed tank, the peristaltic pump was started at 40 rpm to maintain a membrane flux of $60 \text{ L m}^{-2} \text{ h}^{-1}$ (LMH). The start time was recorded when permeate first exited the outlet. The samples were collected, filtered through a $0.22 \mu\text{m}$ filter membrane and immediately quenched with 0.2 mL of MeOH solution for subsequent analysis. After each reaction, DI water was used for backwashing, and the membrane module was dried for subsequent experiments.

2.4. Analytical methods

A scanning electron microscope (SEM) (SIGMA300 Zeiss, Germany) and energy dispersive spectroscopy

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standard curve ($R^2 > 0.999$). The total organic carbon (TOC) removal performance in the two systems was analyzed by a total organic carbon analyzer (TOC-L, Shimadzu, Japan). The degradation products of BPA were analyzed and identified using an ultra-performance liquid chromatograph–mass spectrometer (LC/MS) (Ultimate 3000, Thermo, America). In order to evaluate the toxicity of BPA degradation intermediates, the quantitative structure–activity relationship (QSAR) method was used in this study. The acute toxicity, developmental toxicity, mutagenicity and bioaccumulation of BPA and its degradation intermediates were evaluated using the toxicity assessment software (T.E.S.T.).

2.5. Membrane fouling analysis

The resistance in-series model was used to quantitatively analyze the types of membrane fouling in this study. The intrinsic membrane resistance was calculated based on the transmembrane pressure (TMP) when filtering deionized water. After hydraulic backwashing, membrane fouling can be divided into reversible membrane resistance (R_r) and irreversible membrane resistance (R_{ir}), as shown in [eqn \(1\) and \(2\)](#):

$$\text{[Blank Box]} \quad (1)$$

$$R_f = R_{ir} + R_r \quad (2)$$

where R_t is the total membrane fouling resistance (m^{-1}); TMP is the trans-membrane pressure (Pa); μ is the dynamic viscosity of the feed water (Pa s); J is the permeate flux (LMH); R_m is the intrinsic membrane resistance (m^{-1}); R_f is the total fouling resistance (m^{-1}); R_{ir} is the hydraulic irreversible fouling resistance (m^{-1}); and R_r is the hydraulic reversible fouling resistance (m^{-1}).

During the experiment, the flux was maintained at 60 LMH. First, the TMP was recorded while filtering 100 mL of DI water using a pressure gauge, and the average value was calculated as TMP_0 . The R_m value of the membrane was calculated according to [eqn \(3\)](#). Subsequently, 1200 mL of the water sample was added and operated under the same conditions. When the processed volume reached 500 mL, filtration

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3. Results and discussion

3.1. Process performance of the integrated systems

3.1.1. BPA removal of the integrated systems M_0 and M_{Co-Mn} were assembled to establish baseline and catalytic membrane systems, respectively. The BPA removal performance was evaluated under the same conditions, and the results are illustrated in [Fig. 2a](#). The comparative experiments revealed that M_0 and M_{Co-Mn} exhibited minimal BPA adsorption. Furthermore, PMS alone and the M_0 /PMS system demonstrated very low BPA removal. When PMS was added, the BPA removal rate of M_{Co-Mn} /PMS could reach 85%, indicating that the catalytic membrane could effectively degrade BPA. After coupling GAC, MG_1 and MG_2 could adsorb more than 90% of BPA, so by combining advanced oxidation with GAC adsorption, the MG_1 /PMS and MG_2 /PMS systems could remove 100% of BPA in 5 min.

Fig. 2 (a) BPA removal in different systems and (b) removal efficiency of BPA and TOC in different systems ($[BPA]_0 = 10 \text{ mg L}^{-1}$, $[PMS]_0 = 0.05 \text{ mM}$, $[Flux] = 60 \text{ LMH}$).

The MG_1 /PMS and MG_2 /PMS systems could degrade BPA in a relatively short time, which might be because GAC itself has the function of adsorption and catalysis.²⁹ As shown in Fig. S2, the activation rate of PMS after reaction for 5 min in the three systems indicated that the utilization rate of PMS was not significantly improved after coupling with GAC. Therefore, GAC mainly adsorbed BPA in the

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molecular substances. Therefore, the coupling of activated carbon with a catalytic ceramic membrane can significantly improve the removal rates of BPA and TOC.

3.1.2. Impact of operational parameters A systematic investigation was conducted to evaluate key operational parameters (membrane flux, initial BPA concentration, PMS concentration, pH, and operational temperature) in the M_{Co-Mn}/PMS , MG_1/PMS and MG_2/PMS systems. [Fig. 3a](#) shows the effect of membrane flux on BPA removal in the integrated systems. It was found that the removal rate of BPA decreased slowly with increasing membrane flux. Increasing the membrane flux shortened the contact time between PMS and the catalyst, resulting in fewer active species being produced and a lower removal efficiency of BPA.³⁰ [Fig. 3b](#) shows the effect of the BPA initial concentration on BPA degradation in the three systems. It was observed that with the increase of BPA initial concentration from 2 mg L^{-1} to 20 mg L^{-1} , the BPA removal rates of the M_{Co-Mn}/PMS , MG_1/PMS and MG_2/PMS systems decreased by 20%, 15% and 13%, respectively. This might be due to a decrease in the number of active free radicals obtained by each BPA molecule at higher concentrations of BPA, resulting in a decrease in the efficiency of BPA removal.³¹ The decreases in BPA removal rates in the MG_1/PMS and MG_2/PMS systems were relatively small, primarily attributed to the synergistic adsorption capacity of activated carbon that enabled non-degraded BPA molecules to be removed by activated carbon adsorption.³² The effects of PMS concentration on BPA degradation in the M_{Co-Mn}/PMS , MG_1/PMS and MG_2/PMS systems are shown in [Fig. 3c](#). The BPA removal rates of the M_{Co-Mn}/PMS , MG_1/PMS and MG_2/PMS systems increased by 43%, 29% and 26%, respectively, upon increasing the PMS concentration from 0.01 mM to 0.1 mM. In addition, the effects of initial solution pH on BPA removal by the M_{Co-Mn}/PMS , MG_1/PMS and MG_2/PMS systems were investigated. As shown in [Fig. 3d](#), pH variation exerted a relatively weak influence on the M_{Co-Mn}/PMS , MG_1/PMS and MG_2/PMS systems. Remarkably, the MG_1/PMS and MG_2/PMS systems maintained a BPA removal rate of over 90% even under extreme acidic (pH 3) or alkaline (pH 11) conditions. The temperature-dependent catalytic performance was systematically evaluated, as illustrated in [Fig. 3e](#). When the temperature increased from 15°C to 35°C , the BPA removal rates of the M_{Co-Mn}/PMS , MG_1/PMS and MG_2/PMS systems increased by 8%, 12% and 10%, respectively. Systematic evaluation revealed that temperature changes had little effect on the BPA

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temperature, and the presence of competing inorganic anions. While system performance demonstrated sensitivity to membrane flux, initial BPA concentration, and PMS concentration, the degradation efficiency consistently exceeded 70%, underscoring the robust nature of these hybrid processes. These results further confirmed the stability and reliability of the two combined processes in the treatment of complex water quality, as well as robustness to interference from environmental interference factors.



Fig. 3 Effects of (a) permeate flux, (b) BPA concentration, (c) PMS concentration, (d) pH, (e) operating temperature, and (f) coexisting anions on BPA removal in M_{Co-Mn}/PMS , MG_1/PMS and MG_2/PMS systems ($[BPA]_0 = 10 \text{ mg L}^{-1}$, $[PMS]_0 = 0.1 \text{ mM}$, $[Flux] = 60 \text{ LMH}$, $pH = 7$).

3.2. Analysis of dominant reactive species and stability

In this study, the dominant reactive species responsible for BPA degradation in the MG_1/PMS and MG_2/PMS systems were investigated by a series of free radical quenching experiments. MeOH was used as the capture agent for $\cdot OH$ and $SO_4^{\cdot -}$, TBA as the capture agent for $\cdot OH$, and DMSO as the probe for high-valence metal-oxo species (HVMO).³³ As shown in [Fig. 4a](#), after adding 500 mM MeOH to the MG_1/PMS and MG_2/PMS systems, the degradation rates of BPA decreased by 7.8% and 11.8%, respectively. When 500 mM TBA was added to the two systems, the degradation rates decreased by 2.8% and 4.1%, respectively. The results indicated that $\cdot OH$ and $SO_4^{\cdot -}$ were both generated in the MG_1/PMS and MG_2/PMS systems, but they were not the dominant species responsible for BPA degradation in the two systems. Furthermore, when 10 mM DMSO, as the probe for high valence metal-oxo species, was added to the two systems, the degradation rates significantly decreased by 35.2% and 30.5%, respectively, in the MG_1/PMS and MG_2/PMS systems. The results showed that HVMO played a more significant role in the two systems. Notably, although HVMO played a more important role than $\cdot OH$ and $SO_4^{\cdot -}$, the presence of activated carbon in the systems ensured that the BPA removal rate remained above 60% in both systems even after the quenching reaction. The high-resolution XPS

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Fig. 4 Effects of (a) coexisting anions and (b) long cycle operation on the removal of BPA in different systems ($[BPA]_0 = 10 \text{ mg L}^{-1}$, $[PMS]_0 = 0.1 \text{ mM}$, $[Flux] = 60 \text{ LMH}$, $pH = 7$).

The degradation efficiency of BPA by the M_{Co-Mn}/PMS , MG_1/PMS , and MG_2/PMS systems during long-term operation was compared to evaluate the stability. As shown in [Fig. 4b](#), the single MG_1 and MG_2 systems initially achieved BPA removal through activated carbon adsorption, but the BPA removal efficiency significantly decreased as the reaction progressed due to adsorption saturation. The M_{Co-Mn}/PMS system demonstrated a marked decline in catalytic efficiency, exhibiting an initial BPA removal efficiency of 83% at 30 min that progressively deteriorated to 40.6% after 420 min of continuous operation, suggesting relatively poor stability of BPA removal in the single M_{Co-Mn}/PMS system. In comparison, the MG_1/PMS and MG_2/PMS systems maintained superior operational stability, with initial removal efficiencies of 94.4% and 93.3% at 30 min, respectively, which were only attenuated to 61.2% and 69.1% after 420 min of operation. This long-term stable removal efficiency was attributed to the sustained adsorption of activated carbon and the synergistic effect of ceramic membrane catalytic oxidation, effectively avoiding the problems of adsorption saturation and reduction in catalytic efficiency.^{34,35}

3.3. Analysis of degradation pathway and toxicity

3.3.1. Degradation pathway The degradation intermediate by-products of BPA in the MG_1/PMS and MG_2/PMS systems were elucidated through LC-MS analysis. The results obtained are shown in [Fig. 5](#). As shown in [Fig. 5a](#), there were two potential pathways for the degradation of BPA in the MG_1/PMS system, resulting in 6 possible degradation intermediates. In pathway 1, active species generated by the reaction resulted in the breaking of the C–C bond of BPA to form the intermediate product P1 ($m/z = 150$) connected to a methyl radical. P1 subsequently underwent an oxidation reaction and combined with $\cdot OH$ to generate P2 ($m/z = 152$). In pathway 2, BPA underwent a hydroxylation reaction to generate

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species into smaller fragments P4 ($m/z = 210$) and ultimately degraded into P5 ($m/z = 60$) or mineralized to CO_2 and H_2O .



Fig. 5 BPA degradation pathways of (a) the MG_1/PMS system and (b) the MG_2/PMS system.

The above results show that, in the MG_1/PMS system, the activated carbon was in close contact with the catalytic ceramic membrane, and GAC preferentially adsorbed BPA and its macromolecular intermediate products (such as P1 and P3), accelerating the attack of free radicals on pollutants through a local enrichment effect and inhibiting coupling side reactions (the formation of P3 in pathway 2).³⁹ Compared with the 11 intermediate products produced by the $\text{M}_{\text{Co-Mn}}/\text{PMS}$ system previously studied, the number of intermediate products in the MG_1/PMS system was decreased to 6.²⁷ The degradation pathway of the MG_1/PMS system was significantly shortened, and the TOC removal rate was increased to 66%. In the MG_2/PMS system, as the GAC was located at the back end of the oxidation reaction, oxidation and adsorption proceeded step by step. Oxidative intermediate products (such as P3, $m/z = 362$) underwent intermolecular coupling reactions (pathway 2) to form macromolecular substances. The GAC tube extended the hydraulic retention time, enhanced the contact between GAC and small molecules, effectively adsorbed BPA and its transformation intermediate products, and increased the TOC removal rate to 73% ([Fig. 2b](#)).

3.3.2. Evaluation of potential toxicity This investigation systematically assessed the toxicity of intermediate products generated in the MG_1/PMS and MG_2/PMS systems. The toxicity evaluation was performed using the median lethal concentration (LC_{50}) for fathead minnows, the oral LD_{50} for rats, the LC_{50} for *Daphnia magna* Straus, and the bioaccumulation factor (BF) with quantitative structure–activity relationship (QSAR) modeling predictions.^{40–42} As shown in [Fig. 6](#), the toxicity assessment revealed that the fathead minnow LC_{50} and the *Daphnia magna* Straus LC_{50} of intermediate products produced in both systems were significantly higher than those of BPA. The overall acute toxicity levels

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lower bioaccumulation potential.



Fig. 6 Toxicity variations of (a and e) fathead minnow LC_{50} , (b and f) *Daphnia magna* LC_{50} , (c and g) oral rat LD_{50} and (d and h) bioconcentration factor in the degradation pathways of BPA in the MG_1 /PMS and MG_2 /PMS systems.

3.4. Membrane fouling analysis

A systematic evaluation of membrane fouling behavior was conducted using Dingjiahe Reservoir water as the background raw water, with a particular focus on the relative transmembrane pressure (P/P_0) changes and the reversibility of fouling in the catalytic ceramic membrane system. As shown in [Fig. 7a](#), the M_{Co-Mn} /PMS system exhibited excellent antifouling performance. After three cycles, its final P/P_0 remained at 5.33, while P/P_0 values of the M_0 , M_{Co-Mn} , and M_0 /PMS systems were 9.5, 8.5, and 7.55, respectively. [Fig. 7b](#) shows that the R_f values of M_0 , M_{Co-Mn} , and M_0 /PMS were $1.369 \times 10^{12} \text{ m}^{-1}$, $1.305 \times 10^{12} \text{ m}^{-1}$ and $1.253 \times 10^{12} \text{ m}^{-1}$, respectively. However, the M_{Co-Mn} /PMS system exhibited a substantially reduced R_f of $4.23 \times 10^{11} \text{ m}^{-1}$, representing a reduction of over 65% compared to the other three systems. In the M_{Co-Mn} /PMS system, the Co–Mn catalyst on the ceramic membrane effectively activated PMS to generate reactive species for oxidizing and decomposing large organic molecules, thereby reducing the formation of filter cake layers. In addition, the active species alleviated pore blockage by degrading small-molecule pollutants adsorbed in the membrane pores.



Fig. 7 (a and b) Membrane fouling generated, R_f and R_{ir} by different systems filtering actual water. (c and d) Effects of different PMS concentrations on membrane fouling generated, R_f and R_{ir} under the M_{Co-Mn} /PMS system filtering actual water. (e and f) Effects of different fluxes on membrane fouling generated, R_f and R_{ir} under the M_{Co-Mn} /PMS system filtering actual water ([Flux] = 60 LMH, [PMS] = 0.2 mM).

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appropriate PMS concentration, the M_{Co-Mn}/PMS system could significantly improve the alleviation efficiency of membrane fouling, reduce irreversible fouling resistance, and enhance the long-term stable operational capability of the membrane. Furthermore, Fig. S4 and Table S4 show the fitting analysis of five composite pollution models for the M_0 and M_{Co-Mn}/PMS systems. M_0 exhibited the highest fitting degree with the model of complete blockage of the filter cake layer, $R^2 = 0.9992$. In the M_{Co-Mn}/PMS system, the intermediate standard blockage model became predominant at 0.2 mM PMS, $R^2 = 0.9925$, indicating partial pore constriction without complete occlusion. The addition of PMS significantly enhanced the decomposition of pollutants on the membrane surface, reduced the accumulation of pollutants at the membrane pore orifices, and facilitated deeper foulant penetration into the membrane matrix. Although increasing the PMS concentration to 0.3 mM slightly reduced the fitting quality, $R^2 = 0.9923$, the persistent dominance of the intermediate standard blockage model confirmed the system's stability. The minor fitting degrees of other models remained stable across different PMS concentrations. The M_{Co-Mn}/PMS system optimized the antifouling ability and reduced membrane fouling through precise PMS concentration control.

The effects of different membrane fluxes on membrane fouling in the M_{Co-Mn}/PMS system were investigated and are shown in [Fig. 7e and f](#). At a flux of 40 LMH, R_r and R_{ir} were $0.68 \times 10^{11} \text{ m}^{-1}$ and $1.58 \times 10^{11} \text{ m}^{-1}$, respectively, with a P/P_0 of 2.12. When the flux was 60 LMH, P/P_0 was 2.23 and R_f was $2.5 \times 10^{11} \text{ m}^{-1}$. The M_{Co-Mn}/PMS system exhibited excellent antifouling performance with minimal parameter fluctuations. In contrast, when the flux increased to 80 LMH, after three operating cycles, the P/P_0 value reached 6.73, accompanied by severe membrane fouling. The R_f and R_{ir} values were $4.81 \times 10^{11} \text{ m}^{-1}$ and $3.35 \times 10^{11} \text{ m}^{-1}$, respectively. Under relatively low flux conditions, the contact time between pollutants and active species was prolonged, achieving complete oxidative degradation. Conversely, excessive flux accelerated the migration of pollutants, leading to severe membrane fouling.

The surface characterization through SEM elucidated the formation of actual water membrane surface fouling in membrane filtration of different systems, as shown in [Fig. 8](#). Membrane samples were backwashed with 0.2 mM PMS solution for 30 min and left to stand undisturbed for 24 h after a complete filtration cycle. M_0 developed a dense cake layer with irregular aggregates, demonstrating classical

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and EDS analysis results, it can be seen that the M_{Co-Mn}/PMS system can not only effectively decompose pollutants and reduce membrane fouling, but also maintain the long-term stability and activity of the catalyst.



Fig. 8 SEM images of (a and b) M_0 , (c and d) M_{Co-Mn} and (e and f) M_{Co-Mn}/PMS after filtering actual water.

4. Conclusions

In this study, granular activated carbon (GAC) was loaded into the catalytic ceramic membrane channels (MG_1) and connected at the rear end of the catalytic ceramic membrane (MG_2) for improving the removal of emerging contaminants *via* PMS activation. Compared with the catalytic ceramic membrane system without GAC coupling (M_{Co-Mn}/PMS), the MG_1/PMS and MG_2/PMS systems demonstrated complete removal of BPA within 5 min, with TOC removal rates increasing to 66% and 73%, respectively. The quenching experiments showed that the MG_1/PMS and MG_2/PMS systems utilized hybrid oxidation pathways involving HVMO (Co^{4+}/Mn^{5+}) and $SO_4^{\cdot-}$. LC-MS analysis indicated that degradation occurred through two pathways, and compared with the 11 intermediate products produced by M_{Co-Mn}/PMS , the intermediate products in the MG_1/PMS and MG_2/PMS systems were reduced to 6 and 5, respectively. The comprehensive characterization through LC-MS and toxicity analysis confirmed the effective treatment of low-concentration wastewater and an overall reduction in toxicity. The fitting of the composite pollution model showed that the M_{Co-Mn}/PMS system exhibited significant antifouling ability, and the membrane fouling mechanism was mainly based on intermediate standard blockage. After three consecutive operational cycles, the M_{Co-Mn}/PMS system significantly exhibited a remarkable 69.1% reduction in R_f compared to the M_0 alone, indicating its significant advantage in controlling membrane fouling. Therefore, the construction of the MG_1/PMS and MG_2/PMS systems provides a research direction for the efficient removal of emerging contaminants and the deep purification of water.

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Conflicts of interest

There are no conflicts of interest to declare.

Data availability

The data supporting the findings of this study are available within the paper and its supplementary information (SI). Supplementary information is available. See DOI: <https://doi.org/10.1039/d5nr03774c>.

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